

Supply Chain Analytics

An End-to-End Analytics Project

Excel • Python • SQL • Power BI

Dataset: DataCo Smart Supply Chain (2015–2018)

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Executive Summary

This is a self-directed analytics project built on a supply chain and e-commerce dataset covering orders, shipping, products, categories, and customers between 2015 and early 2018. The goal was to practice a complete, real-world analyst workflow rather than a single-tool exercise: extracting and deduplicating data in Excel, cleaning and standardizing it in Python, writing business-focused SQL queries, and building an executive-style Power BI dashboard powered by DAX. Twelve SQL queries were written to answer real business questions such as delivery performance, product and category profitability, customer repeat behavior, and discount impact on profit. These findings were translated into a three-page Power BI dashboard (Executive Dashboard, Sales Analysis, and Profitability & Delivery Analysis). The analysis surfaced several concrete insights, including a very low on-time delivery rate that needs review, strong concentration of revenue in a few products and categories, and clear evidence that heavy discounting reduces profit in some categories.

Project Objective

- Practice the full analyst workflow end to end: Excel to Python to SQL to Power BI, on one connected dataset.
- Clean and prepare a multi-table supply chain dataset (orders, shipping, products, categories, customers) for analysis.
- Write medium-to-advanced SQL queries to answer real business questions, not just basic SELECT statements.
- Design and build a multi-page, executive-style Power BI dashboard using DAX measures and interactive visuals.
- Turn the technical analysis into clear, non-technical business insights and recommendations.

Dataset Overview

The project uses five related tables from the DataCo Smart Supply Chain dataset:

Table	Description
orders	Order-line level data: order date, market, region, sales, profit, discount, and customer/product IDs. ~180,519 rows.
shipping	One record per order: shipping mode, delivery status, scheduled vs. actual shipping days, late-delivery risk flag.
products	Product catalog: product name, category ID, department, and price.
category	Category ID mapped to category name and department name.
customer	Customer ID mapped to name, city, state, and country.

The data spans January 2015 to January 2018 and covers 5 global markets (Pacific Asia, USCA, Africa, Europe, LATAM).

Data Preparation (Excel & Python)

Excel — Extraction & Deduplication

- Extracted the raw tables (orders, shipping, products, category, customers) from the original source data into separate worksheets.
- Identified and removed duplicate rows in the extracted tables before handing the data off to Python for further cleaning.

Python — Cleaning, Feature Prep & Load to SQL

Cleaning and preprocessing was carried out in Python using pandas, then the cleaned tables were loaded directly into MySQL for the SQL analysis stage.

Loaded all five tables and checked structure

Each CSV (customer, Orders, category, product, shipping) was loaded with pandas, then checked with `.info()`, `.shape`, and `.isnull().sum() / .shape[0] * 100` to see column data types, row counts, and the percentage of missing values in every column before deciding how to handle them.

Rounded currency and price columns

Total Sales, Benefit per order, Order Item Discount, price per item, and Order Item Total were all rounded to 2 decimal places for consistency, since raw values had long floating-point tails.

Dropped unused columns

Latitude, Longitude, Order Item Profit Ratio, and Order Profit Per Order were dropped from the orders table, since they were not needed for this analysis and reduced file size.

Renamed columns for clarity

'order date (DateOrders)' was renamed to 'order date', 'Order Item Product Price' to 'price per item', and 'Sales' to 'Total Sales' so column names read clearly in SQL and Power BI.

Fixed mixed date formats

The order date column contained two different date formats in the same column (e.g. 1/31/2018 and 31-01-2018). This was standardized using `pandas.to_datetime(..., format="mixed")` so every row could be read automatically regardless of its original format, then the time portion was removed to leave a clean date.

Loaded cleaned tables into MySQL

Using SQLAlchemy and PyMySQL, all five cleaned tables were pushed directly into a MySQL database (supply_data) with `to_sql()`, making them ready for the SQL analysis stage without any manual import.

SQL Analysis

Twelve SQL queries were written against the dataset, covering joins, GROUP BY, HAVING-style filtering, CASE WHEN segmentation, window functions (RANK, LAG), and subqueries. Each query, its purpose, and the insight it provides are listed below.

Q1. Markets Above Minimum Order Volume

Purpose: Shows total orders, revenue, and profit for each market, so leadership can see which markets drive the most business.

```
select Market, count(distinct `Order Id`) as total_orders,
       SUM(`Total Sales`) AS total_revenue,
       SUM(`Benefit per order`) AS total_profit
from orders
group by Market
order by total_revenue DESC;
```

Insight: Identifies the highest-revenue and highest-profit markets, helping prioritize where sales and marketing effort should be focused.

Q2. On-Time vs Late Delivery Rate by Shipping Mode

Purpose: Calculates the percentage of on-time, late, and advance deliveries for each shipping mode.

```
SELECT `Shipping Mode`,
       ROUND(100.0 * SUM(CASE WHEN `Delivery Status` = 'Shipping on time' THEN 1 ELSE 0 END) / COUNT(*),
2) AS on_time_pct,
       ROUND(100.0 * SUM(CASE WHEN `Delivery Status` = 'Late delivery' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS late_delivery_pct,
       ROUND(100.0 * SUM(CASE WHEN `Delivery Status` = 'Advance shipping' THEN 1 ELSE 0 END) / COUNT(*),
2) AS advance_delivery_pct
```

```
FROM shipping
GROUP BY `Shipping Mode`
ORDER BY on_time_pct DESC;
```

Insight: Reveals which shipping modes are the most and least reliable, supporting decisions on carrier selection and delivery promises.

Q3. Average Delivery Delay by Market

Purpose: Finds the average number of days a shipment is late (or early) compared to the promised delivery date, for each market.

```
select o.Market,
round(avg(s.`Days for shipping (real)` - s.`Days for shipment (scheduled)`),2) as avg_delay_days
from orders o inner join shipping s
on s.`Order ID` = o.`Order Id`
group by o.Market
order by avg_delay_days;
```

Insight: Highlights which markets have the biggest gap between promised and actual delivery time, pointing to logistics issues by region.

Q4. Monthly Sales Trend

Purpose: Shows total sales for each calendar month, to see seasonal patterns.

```
select monthname(`order date`) as month,
round(sum(`Total Sales`),2) as monthly_revenue
from orders
group by month
order by monthly_revenue desc;
```

Insight: Useful for spotting which months are strongest for sales. Note: because the dataset spans multiple years (2015-2018), grouping by month name alone combines all years together; adding a Year column would give a cleaner year-by-year trend.

Q5. Month-over-Month Sales Growth

Purpose: Calculates how much sales grew or declined compared to the previous month, using the LAG window function.

```
with monthly_sales as(
select month(`order date`) as month, YEAR(`order date`) AS yr,
round(sum(`Total Sales`),2) as Total_sales
from orders
group by month, yr
)
select month, yr, Total_sales,
lag(Total_sales) over(order by month) as prev_month_rev,
round(((Total_sales - lag(Total_sales) over(order by month)) / lag(Total_sales) over(order by month))*100,2) AS month_growth_pct
from monthly_sales
order by yr, month;
```

Insight: Shows the direction (growth or decline) of sales month to month. For a fully accurate chronological sequence, the LAG function should order by both year and month together.

Q6. Top 10 Products by Revenue

Purpose: Lists the 10 products that generated the highest total sales.

```
select p.`Product Name`, round(sum(o.`Total sales`),2) as total_revenue
from orders o join products p
on o.`Product Card Id` = p.`Product Card Id`
```

```
group by p.`Product Name`
order by total_revenue desc
limit 10;
```

Insight: Identifies the best-selling products, which should get priority in marketing, stocking, and promotions.

Q7. Category Profitability

Purpose: Shows total sales, total profit, and profit margin percentage for every product category.

```
select c.`Category Name`,
round(sum(o.`Total Sales`),2) as total_sales,
round(sum(o.`Benefit per order`),2) as total_profit,
round((sum(o.`Benefit per order`)/sum(o.`Total Sales`))*100,2) as total_margin_perct
from orders o left join products p on p.`Product Card Id` = o.`Product Card Id`
join category c on c.`Product Category Id` = p.`Product Category Id`
group by c.`Category Name`
order by total_sales, total_profit, total_margin_perct;
```

Insight: Shows that high revenue does not always mean high profit — some categories sell a lot but earn thin margins, while others are smaller but more profitable.

Q8. Rank Products Within Each Category

Purpose: Finds the best-selling product inside every category using the RANK window function.

```
SELECT `Category Name`, `Product Name`, total_sales,
RANK() OVER (PARTITION BY `Category Name` ORDER BY total_sales DESC) AS product_rank
FROM (
    SELECT c.`Category Name`, p.`Product Name`, SUM(o.`Total Sales`) AS total_sales
    FROM orders o
    left JOIN products p ON o.`Product Card Id` = p.`Product Card Id`
    JOIN category c ON p.`Product Category Id` = c.`Product Category Id`
    GROUP BY c.`Category Name`, p.`Product Name`
) AS t
ORDER BY `Category Name`, product_rank;
```

Insight: Gives a clear 'best in category' list instead of one long mixed ranking, useful for category managers.

Q9. Repeat vs One-Time Customers

Purpose: Counts how many customers placed only one order versus customers who ordered more than once.

```
WITH customer_orders AS (
    SELECT `Customer Id`, COUNT(DISTINCT `Order Id`) AS order_count
    FROM orders GROUP BY `Customer Id`
)
SELECT CASE WHEN order_count = 1 THEN 'One-Time Customer' ELSE 'Repeat Customer' END AS
customer_type,
COUNT(*) AS num_customers
FROM customer_orders
GROUP BY CASE WHEN order_count = 1 THEN 'One-Time Customer' ELSE 'Repeat Customer' END;
```

Insight: Shows what portion of the customer base is repeat business versus one-time buyers — a direct indicator of customer retention.

Q10. Discount Rate vs. Average Profit

Purpose: Groups orders into discount bands (No Discount, 1-10%, 11-20%, 21%+) and compares average profit in each band.

```
SELECT
CASE
    WHEN `Order Item Discount Rate` = 0 THEN 'No Discount'
    WHEN `Order Item Discount Rate` <= 0.10 THEN '1-10%'
    WHEN `Order Item Discount Rate` > 0.10 AND <= 0.20 THEN '11-20%'
    WHEN `Order Item Discount Rate` > 0.20 THEN '21%+'
END AS discount_band,
```

```

        WHEN `Order Item Discount Rate` <= 0.20 THEN '11-20%'
        ELSE '21%+'
    END AS discount_band,
    COUNT(*) AS order_lines,
    ROUND(AVG(`Benefit per order`), 2) AS avg_profit
FROM orders
GROUP BY discount_band
ORDER BY discount_band;

```

Insight: Tests whether bigger discounts are hurting profit. This directly supports the 'Profit Margin % and Discount % by Month' chart on the dashboard.

Q11. Cancelled Orders by Region

Purpose: Finds only the regions whose order cancellation rate is above the overall average cancellation rate, using a subquery.

```

WITH region_cancel_rate AS (
    SELECT `Order Region`,
        ROUND(100.0 * SUM(CASE WHEN `Order Status` = 'CANCELED' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
cancel_rate_pct
    FROM orders GROUP BY `Order Region`
)
SELECT * FROM region_cancel_rate
WHERE cancel_rate_pct > (SELECT AVG(cancel_rate_pct) FROM region_cancel_rate)
ORDER BY cancel_rate_pct DESC;

```

Insight: Flags specific regions with unusually high cancellation rates, worth investigating for service or fraud issues.

Q12. Orders Missing Shipping Records

Purpose: A data-quality check that finds any order with no matching shipping record, using a LEFT JOIN.

```

SELECT o.`Order Id`, o.`Market`, s.`Delivery Status`
FROM orders o
LEFT JOIN shipping s ON o.`Order Id` = s.`Order Id`
WHERE s.`Order Id` is null;

```

Insight: Confirms whether every order has a matching shipping record. If this query returns rows, it means some orders have missing delivery data that should be investigated before trusting delivery KPIs.

Dashboard Analysis

The Power BI report is built as three pages: Executive Dashboard, Sales Analysis, and Profitability & Delivery Analysis. Slicers for Market, Order Region, and Category Name are available on every page for consistent filtering.

Page 1 — Executive Dashboard

- Total Sales / Total Orders / Total Profit / On-Time % Delivery (KPI cards): The four headline numbers for the whole business, giving an at-a-glance health check.
- Monthly Sales Trend (line chart): Shows how total sales move month to month, useful for spotting seasonality.
- Sales vs Late Delivery % (combo chart): Overlays sales volume with late-delivery risk percentage by month, showing whether delivery problems track with demand.
- Sales Quantity vs Profit by Product (scatter chart): Plots each product by quantity sold against profit generated, highlighting high-volume, high-profit products.
- Sales by Category (bar chart): Ranks categories by total sales, showing Fishing as the clear leader.
- Category Sales vs Late Delivery Risk (combo chart): Same combo-chart idea as above, but broken down by category instead of by month.

- Top 10 Products (table): Lists the ten highest-revenue products by name and total sales.

Page 2 — Sales Analysis

- Total Sales by Year-Month and Category (multi-line chart): Tracks each category's sales trend over time on one chart, making it easy to compare categories side by side.
- Category-Product Matrix (heatmap table): An expandable Category → Product breakdown of sales by month, with row and column totals, giving month-level detail behind the summary charts.

Page 3 — Profitability & Delivery Analysis

- Profit Margin % and Discount % by Month (combo chart): Compares average discount rate against profit margin over time to check whether discounting is eroding profit.
- Net Order Volume Trend (area chart): Tracks the total number of orders placed each month.
- Orders by Delivery Status (bar chart): Breaks down all orders into Late delivery, Advance shipping, Shipping on time, and Shipping canceled.
- Category x Month → Profit Margin % (matrix): Shows the exact profit margin for every category, in every month.
- Category x Month → Avg Discount % (matrix): Shows the exact average discount rate for every category, in every month, matched to the margin matrix above.

DAX Measures

The following DAX measures power the KPI cards and charts across all three dashboard pages. Note: the exact formula text stored inside the Power BI (.pbix) data model could not be extracted automatically, since Power BI stores it in a compressed binary format. The formulas below reflect the measures specified and used while building this dashboard, and should be checked against the live report for an exact match.

Total Sales

```
Total Sales =
SUM(orders[Total Sales])
```

Adds up total revenue across all orders.

Total Orders

```
Total Orders =
DISTINCTCOUNT(orders[Order Id])
```

Counts the number of unique orders.

Total Profit

```
Total Profit =
SUM(orders[Benefit per order])
```

Adds up total profit across all orders.

Profit Margin %

```
profit Margin = (sum('supply_data orders'[Benefit per order])/sum('supply_data orders'[Total Sales]))*100
```

Profit as a percentage of sales.

On-Time Delivery Rate


```
on time % delivery = (CALCULATE(countrows('supply_data shipping'),'supply_data shipping'[Delivery Status]="Shipping on time")/COUNTROWS('supply_data orders'))*100
```

Percentage of orders delivered on time.

Late Delivery Rate

```
Late Delivery Risk (in%) = (sum('supply_data shipping'[Late_delivery_risk])/countrows('supply_data shipping'))*100
```

Percentage of orders flagged as a late-delivery risk.

Avg Discount %

```
avg item discount in % = AVERAGE('supply_data orders'[Order Item Discount Rate]) * 100
```

Average discount rate applied, shown as a percentage.

Avg Order Value

```
Avg Order Value =  
DIVIDE([Total Sales], [Total Orders], 0)
```

Average revenue per order.

Key Business Insights

- The Fishing category is by far the top-performing category, generating close to 6.9M in sales — far ahead of every other category.
- The single best-selling product (Field & Stream Sportsman 16 Gun Fire Safe) alone accounts for roughly 6.93M in sales, making it the most important SKU in the catalog.
- The dashboard's on-time delivery KPI shows only 17.84%, meaning over 80% of orders are not marked as on-time. This is an unusually low number and should be re-checked against the measure logic before being reported as final.
- Late Delivery Risk stays consistently high, in the 54%-56% range across almost every month, indicating a systemic delivery problem rather than a one-time issue.
- Certain regions show a cancellation rate clearly above the company-wide average (Q11), suggesting localized fulfillment or fraud-related issues worth investigating further.
- Some categories, such as 'As Seen on TV!', show large swings in profit margin from month to month (from +37% down to -20%), suggesting inconsistent pricing or discounting practices.
- A number of product categories in the catalog (Baby, Basketball, Books, Cameras, CDs) show no recorded sales activity at all, meaning the active product range is smaller than the full catalog suggests.
- The customer base includes a meaningful share of one-time buyers (Q9), representing a clear opportunity to improve repeat-purchase / retention marketing.
- Net order volume drops sharply in the final months of the dataset (October-December), which lines up with the dataset simply ending in early 2018 rather than an actual business decline.
- Shipping mode comparison (Q2) shows a measurable gap between the best and worst performing shipping modes on the on-time delivery percentage, useful for renegotiating carrier SLAs.